HRISHIKESH BELAGALI

Undergraduate Research Assistant

Personal Website github.com/hkbelagali
belagal1@msu.edu /in/hkbel

SUMMARY

PhD applicant with research interests in quantum algorithms for time evolution, *ab initio* and machine-learning molecular dynamics, fault-tolerant quantum compilation, and quantum error correction.

SKILLS

Languages: Python, Rust, C++, Tcl, LaTeX

Packages: Qiskit, Cirq, NumPy, PyTorch, JAX, SciPy

Software: LAMMPS, OVITO, NAMD, VMD

EDUCATION

Computer Science (B.S.) — **Michigan State University Honors College**
2024–2027 (expected)
Current GPA: 4.0

RELEVANT COURSEWORK

Quantum Algorithms†, Tensor Networks†, Quantum Physics, Computational Foundations of Artificial Intelligence†, Linear Algebra

†**Graduate course**

EXPERIENCE

2026–Present **Undergraduate Research Assistant** — **MSU-Q** (Advisor: Prof. Ryan LaRose)

- Developed fermionic backpropagation, an efficient classical method for simulating large-scale unitary cluster Jastrow circuits, combining new analytical reductions with a GPU-accelerated implementation to achieve a 70-fold speedup for circuits with up to 77 qubits and 3,500 two-qubit gates.
- Built *propaq*, a Heisenberg propagation package achieving runtimes competitive with state-of-the-art simulators while offering greater customizability, enabling circuit simulations in arbitrary bases with support for custom noise and truncation policies.
- Investigated unitary basis transformations for improving error correction code performance in the presence of biased noise.
- Analyzed the use of the canonical polyadic decomposition for pre-processing chemistry Hamiltonians to reduce time evolution costs.

2025–2026 **Honors Research Scholar** — **MSU Honors College/MSU-Q** (Advisor: Prof. Ryan LaRose)

- Developed a gateset-rotation algorithm for identifying unitary basis transformations to reduce non-Clifford synthesis cost for fault-tolerant quantum algorithms.
- Investigated the utility of unitary basis transformations for Hamiltonian pre-processing in time evolution algorithms.

2024–Present **Undergraduate Research Assistant** — **Vermaas Lab, Plant Research Laboratory** (Advisor: Prof. Josh Vermaas)

- Simulated the effects of butanol and isobutanol on *Clostridium acetobutylicum* and *Zymomonas mobilis* membranes to characterize their impact on membrane dynamics.
- Ran large-scale molecular dynamics simulations on high-performance computing clusters (40,000+ GPU hours and 30TB+ of data) using NAMD.
- Identified key membrane components contributing to alcohol tolerance through data analysis in Python, Tcl, and VMD.
- Investigated the effect of the PDLP5 protein cap on water transport in PIP1;4 aquaporins via grid-steered molecular dynamics.

PUBLICATIONS

[1] **Hrishikesh Belagali**, Ryan LaRose, "Propaq: A Python package for Heisenberg Propagation," *arXiv:2609.07730 (2026)*. Submitted to *ACM Transactions on Software Engineering and Methodology*.

[2] **Hrishikesh Belagali**, Thomas Van Camp, R. Pradeep, Sourin Das, Namit Anand, Ryan LaRose, "Efficient classical simulation of large-scale unitary cluster Jastrow circuits," *arXiv:2607.21337 (2026)*.
Submitted to *Physical Review Letters* (revised manuscript under review).

[3] **Hrishikesh Belagali**, Nitin Kumar Singh, Josh V. Vermaas, "The Nanoscale Impact of Cyclopropane Fatty Acids and Hopanoids on Alcohol Tolerance in *Clostridium acetobutylicum* and *Zymomonas mobilis*," ChemRxiv:15004463 (2026).
Submitted to *The Journal of Physical Chemistry B* (under revision).

CONFERENCES

2026 *ACS, IL*
Molecular mechanisms of alcohol tolerance in *Clostridium acetobutylicum* and *Zymomonas mobilis*: The role of cyclopropane fatty acids and hopanoids acs.digitellinc.com
Poster presented at ACS Fall 2026 Conference.

2026 *MQC, IN*
Efficient classical simulation of large-scale unitary cluster Jastrow circuits
Poster presented at MQC Entanglement 2026.

2026 *GLBRC, WI*
The Nanoscale Impact of Cyclopropane Fatty Acids and Hopanoids on Alcohol Tolerance in *Clostridium acetobutylicum* and *Zymomonas mobilis*
Poster presented at Great Lakes Bioenergy Research Center Annual Science Meeting.

2026 *MSU UURAF*
Reducing T Gate Counts via Gateset Rotation
Poster presented at MSU Undergraduate Research and Arts Forum. **Awarded first place** in the Engineering, Computer Science and Mathematics division.

2025 *ACS, D.C.*
Aquaporin Water Transit Modulation via Protein Capping acs.digitellinc.com
Poster presented at ACS Fall 2025 Conference.

2025 *MI*
Evaluating the Interaction of Chloroplast-targeting Protein (At4g16060) with Thylakoid Membrane
Poster presented at First Great Lakes Plant Science Conference.

RESEARCH SOFTWARE & OPEN SOURCE

propaq
Python, Rust
propaq github.com/pypi.org
Creator and primary developer of *propaq*, a Python package for Heisenberg propagation, designed for large-scale simulation on HPC clusters.
- Implemented thread-safe kernels and compact representation of Pauli and Majorana strings in Rust for efficient multi-threaded simulation.
- Developed a Python interface integrating *propaq* with Qiskit and Cirq for seamless simulation workflows using PyO3 bindings.
- Designed a plugin system for developing custom noise and truncation policies in Python, Rust, C, and Julia to enable user customization with minimal overhead.

ffsim
Python, Rust
ffsim github.com
Contributed to *ffsim*, a Python package for simulating fermionic quantum systems.
- Implemented fermionic backpropagation from "Efficient classical simulation of large-scale unitary cluster Jastrow circuits" (arXiv:2607.21337) into *ffsim* for fast unitary cluster Jastrow ansatz simulation.
- Created a JAX-based implementation for GPU-accelerated variational optimization.

QuTiP
Python
Quantum Toolbox in Python (QuTiP) github.com
Contributed to QuTiP, an open-source library for simulating quantum systems.
- Implemented functionality to return eigenkets as an operator in the Qobj class.

aqua-blue
Python
aqua-blue github.com/pypi.org
Secondary maintainer of aqua-blue, a Python package for building echo state networks.
- Implemented reservoir state time-lag, sparsity, and spectral radius to improve prediction accuracy.
- Introduced timezone-aware datetime support for seamless interfacing with data pipelines.
- Demonstrated forecasting capability on the quantum time evolution of Gaussian wave packets.

tce-lib
Python
tce-lib github.com
Contributed to tce-lib, a Python package for building tensorial cluster expansion models for concentrated alloys.
- Implemented lazy evaluation of Monte Carlo trajectories for efficient simulation on HPC clusters.

HONORS AND AWARDS

- First place at MSU UURAF in the Engineering, Computer Science and Mathematics division (2026)
- Wielenga Research Scholarship, Honors College (2025–2026) – Awarded to 15–20 MSU students per year
- Dean's List (Fall 2024, Spring 2025, Fall 2025, Spring 2026)
- Senator Joe and Mary Conroy Endowed Scholarship, College of Engineering (2024) – $2,000
- MSU President's Scholarship (2024) – $30,000

RESEARCH COMMUNICATION

2026
MSU CoE, MI

"Student View: Challenging the Quantum Advantage Boundary" **engineering.msu.edu**
Article highlighting my quantum computing research journey, published by the College of Engineering, Michigan State University.

SELECTED PROJECTS

Python
Qiskit

Hamiltonian Tensor Compression for Time Evolution on Quantum Computers
Extended my work on the canonical polyadic decomposition for factorizing chemistry Hamiltonians as part of my Graduate Tensor Networks course taught by Prof. Ryan LaRose.

Python
Qiskit

Adiabatic Quantum Computation **github.com**
Implemented a quantum adiabatic algorithm to solve QUBO-type problems using the Trotter-Suzuki approximation and quantum Ising model formulation techniques.
Extended this project in my Graduate Quantum Algorithms course to investigate the effects of hardware noise on accuracy and convergence.

Python
OVITO
PyMOL

Genetic Annealing to Determine Protein Structures **github.com**
Monte Carlo Genetic Algorithm to determine protein structures through the optimization of Irbäck's off-lattice model energy equation (RMSD < 3.0 Å).

Python

Quadratic Unconstrained Binary Optimization **github.com**
Stochastic tunneling-enhanced simulated annealing for solving QUBO-type problems.